NGUYEN MINH TUAN ANH

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EDUCATION

VNU-HCM High School for the Gifted – Specialized in Physics | *Ho Chi Minh City, Vietnam* **2024 – 2027**
Tran Dai Nghia High School for the Gifted – Specialized in Mathematics | *Ho Chi Minh City, Vietnam* **2020 – 2024**
GPA Grade 9: 9.4/10 **GPA Grade 10:** 9.4/10 **GPA Grade 11:** 9.3/10

STANDARDIZED TEST SCORES

SAT: 1520 (*730 Verbal, 790 Math*) **IELTS: 7.5**
Advanced Placement: Calculus BC: 5/5 (AB subscore: 5/5) Physics C: Mechanics: 4/5

HONORS & AWARDS

4th Place | National | **Innovation** – RMIT Tech Camp 2026 Grand Finals (Scam Shield), **Top 4 of 100 teams nationwide** **2026**
Gold Medal | International | **Physics** – International Math and Science Competitions (STEMCO), Top 5% of 10,000+ **2025**
Silver Award | International | **Mathematics** – Siam International Math and Science Olympics (SIMSO) **2025**
Gold Award | National | **Mathematics** – Siam International Math and Science Olympics (SIMSO) **2025**
Silver Medal | National | **Mathematics** – Asian Science and Mathematics Olympiad (ASMO), Top 12% nationally **2025**
Silver Medal | National | **Mathematics** – Einstein Olympiad (ESO), Top 15% nationally **2026**

APPLIED MATHEMATICS & MODELING PROJECTS

Founder | JointGuard AI | GitHub | Project Page **Jan 2026 – Present**
An open-source simulation study of whether control signals alone can reveal rising joint friction in robot arms; advised by Dr. Nguyen Vinh Hao (HCMUT) and Mr. Phung Van Hung (VNU-HCM High School for the Gifted)

- Built a **physics-based robot-arm simulator** and fault-injection framework to test whether torque and tracking error alone can reveal rising joint friction.
- Trained a Random Forest detector under a leakage-free protocol: caught **32/32 faulty test runs** (0.21 false alarms/min) vs. 16/32 for a baseline; identified the faulty joint in **30/32**.
- Reported failure on held-out heavier, faster conditions (3.9 false alarms/min); released as **open source (MIT)** with reproducible code, automated tests, a technical report and an interactive demo.

Developer | SkyScore | GitHub | Website **Feb 2025 – Present**
An open-source observing planner that finds the best hours to observe a target; advised by Dr. Nguyen Ngoc Duy (VNU-HCM High School for the Gifted); used by USAC and PAsC to plan 2+ observing trips and the HELIOS astronomy workshop

- Scored every hour on four criteria (target altitude, moonlight, cloud cover, atmospheric seeing), computing Sun, Moon and target positions with **Astropy** and pulling hourly cloud forecasts from the **Open-Meteo API**.
- Framed planning as **multi-objective optimization**: best consecutive-hour window search, **Pareto-front** trade-off analysis without fixed weights, and robustness checks over **500 random weight sets**.
- Trained a **Random Forest seeing predictor** (cross-validated MAE, R^2), stating openly that it learns from physics-based proxy labels rather than real measurements; shipped a bilingual (EN/VI) Streamlit app with pytest unit tests.

Co-Founder & Lead Developer | Scam Shield | GitHub | Video Demo | Website **Oct 2025 – Present**
An AI-powered tool that helps users recognize online scams; 4th of 100 teams nationwide at RMIT Tech Camp 2026 (partners: Google, AWS)

- Led development in a **4-member team**, building a **bilingual (VI/EN)** web app and browser extension combining machine-learning models with rule-based checks to flag suspicious text, images, videos and files.
- Reached **100+ active users** across schools and coding competitions; used the tool with The Promethean Flame to teach **50+ rural students** to recognize scam signals.

Orbital Computation Lead | AstroSphere | USAC Astronomy Club | Web App (Vietnamese) **Feb 2026 – Present**
A browser-based 3D celestial-sphere simulator for teaching basic astronomy

- Led the team computing **sidereal time (IAU 1982 conventions)** and equatorial–horizontal coordinate transformations, showing how the sky's apparent motion depends on latitude.
- Validated every coordinate transformation automatically against the astronomy-engine library (**error < 0.1°**); documented the model's limits (J2000.0 epoch; no precession, nutation or refraction corrections).

Leader | Custom D70F700 Refractor Telescope | PAsC Astronomy Club | Product & Observations **Dec 2025 – Present**

- Designed and built a **70 mm, f/10 achromatic refractor**, calculating optical lengths to fabricate the PVC tube assembly and limit chromatic aberration.
- Reached **20x magnification** with available components and a mount borrowed from another club; served as the main instrument at PAsC events, letting **300+ attendees** safely view the Sun through a solar filter.

INTERNSHIP & LAB EXPERIENCE

Engineering Intern | HCMUT Control & Automation Lab | Website | Certificate **May 2025 – Aug 2025**
Ho Chi Minh City University of Technology; advised by Dr. Nguyen Vinh Hao

- Built a **C++/Python GUI** for industrial robot control, improving real-time joint telemetry streaming and UI latency.
- Modeled control loops and robot dynamics in **MATLAB/Simulink**; programmed **ROS-based mobile robots** (Arduino, sensor fusion) and tested obstacle avoidance and motor response on hardware.
- Work with industrial arms motivated the independent project JointGuard AI (see above).

Project Intern | TA Communications | Certificate **Jun 2026 – Jul 2026**
Supervised by Ms. Anna Thuy, Sales & Operations Manager

- Analyzed store-level sales datasets for **three international consumer brands** in Excel and Python, quantifying growth rates, variability and trends across performance distributions.
- Built **dashboards** turning raw sales data into statistical summaries for comparing stores.

Physics Lab Assistant | VNU-HCM High School for the Gifted | Lab Reports **Jan 2026 – May 2026**
Advised by M.Sc. Trinh Thu Thuy

- Calibrated optics and oscillation apparatus; wrote **LaTeX lab reports** with error analysis.

LEADERSHIP

Co-Founder & Head of Academic Department | The Promethean Flame | Fan page **Jun 2025 – Mar 2026**
A student-run tutoring project bringing STEM support to underserved students

- Grew a **40+ member** tutoring club (1,600+ page followers) and led a 10-member team teaching Math, Physics and Chemistry to **50+ students**, including a one-week teaching trip to Dong Thap with MathMarvel.
- Collaborated with "Ngôi Nhà Trí Tuệ" to tutor **70+ students** in storm- and flood-affected areas, helping **85%** keep up with coursework; supported development of its learning platform EDTOP; featured in Thanh Nien Newspaper.

Vice President → Project Lead | USAC Astronomy Club | Fan page **Jan 2025 – Present**
University of Science astronomy club with 24,000+ followers; Vice President (Jan – Sep 2025), Project Lead (Sep 2025 – Present)

- Coordinated talk shows, star- and Moon-observing nights and school exchanges; helped run booths at the **2025 Space Science Festival** (solar observing for **hundreds of visitors**) and **2025 Student–Enterprise Day** (alongside **33 companies**).
- Supported the club's academic articles and shared telescopes and mounts with PAsC; now lead small product projects, including AstroSphere's orbital computation team.

Vice President | PAsC – Astronomy Club, VNU-HCM High School for the Gifted | Fan page **Sep 2025 – Present**

- Managed a **40-member club**; led a 15-student team writing astronomy research articles and organized the HELIOS workshop for **40+ high schoolers**.
- Co-directed "The SENSUS" STEM event with USAC and Tran Dai Nghia's Physics Club for **100+ attendees**, judged by Dr. Tran Hai Cat.
- Ran an interactive astronomy workshop at Math Open Day (Ho Chi Minh City & Da Lat) for **200+ attendees**; raised **7,000,000 VND (~US$270)** in sponsorships for club operations and prototypes.

Co-Founder | Solar Pulse Hub | Website **Apr 2026 – Present**

- Raised **15,000,000 VND (~US$570)** by collecting e-waste for recycling; taught 6+ sessions on kinetic and solar energy to **40+ children** at Tam Duc Shelter.
- Donated 7 solar lights and supplies (**~US$420**) plus **40 scholarships** to Tam Duc Shelter and Suoi Tuong Village.

President | MathMarvel | Fan page **Jan 2025 – Present**

- Ran "SENSTEM," a STEM outreach campaign in Dong Thap with a hands-on science fair (water rockets, RC cars).
- Organized English communication classes for **100 students** and distributed **28,000,000 VND (~US$1,080)** in aid, including 30 health insurance scholarships and 100+ school supply packages.

VOLUNTEER WORK

Tree-Planting Donor | DalaGroup, Da Lat | Website | Certificate **Sep 2026 – Oct 2026**
Teaching Volunteer | Sugar Vietnam – Duong Den Tri Thuc | Fan page | Certificate **Feb 2025 – May 2025**
Volunteer | Supply distribution to 300+ ethnic minority families, Chu Puh District, Gia Lai | Website | Images **Aug 2024**

ONLINE COURSEWORK

Machine Learning Specialization | DeepLearning.AI & Stanford Online (Coursera) **2026**
Mathematics for Machine Learning Specialization | Imperial College London (Coursera) **2026**

SKILLS & INTERESTS

Engineering & Math: Robot kinematics & dynamics, control systems, ROS, time-series analysis, multi-objective optimization
Programming & Tools: Python (scikit-learn, Astropy, Streamlit), C++, Java, MATLAB/Simulink, PyBullet, Arduino, Git
Languages: Vietnamese (native), English (fluent, IELTS 7.5) **Interests:** Chinese chess, soccer, table tennis